Pressure-coupling qualification withdrawn

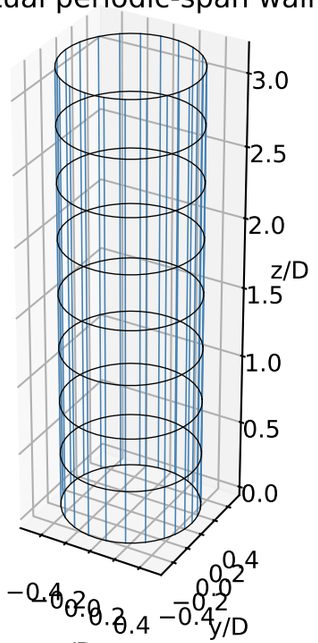
These historical startup fields contain mesh-locked pressure modes.

Native face continuity and the momentum ledger passed, but did not verify pressure / cell-velocity coupling. Both numerical and physical qualification remain false. Original numerical evidence is preserved.

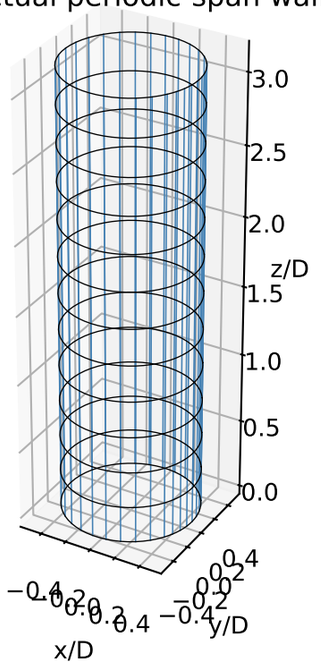
See verification-cylinder3900-pressure-coupling/report.md for the correction, unsmoothed old/new fields and matched-time dt refinement.

Native checkpoint geometry: body-fitted cylinder, not a Cartesian mask

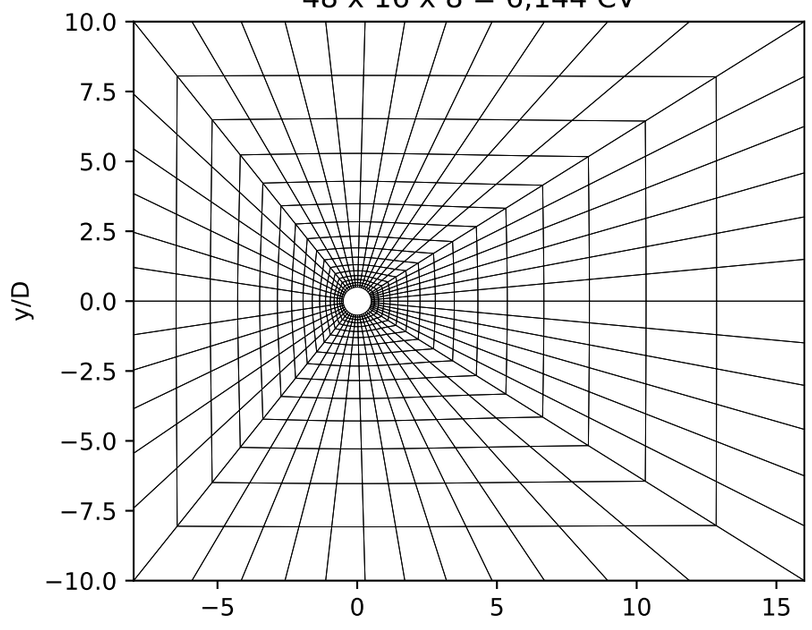
coarse: actual periodic-span wall mesh



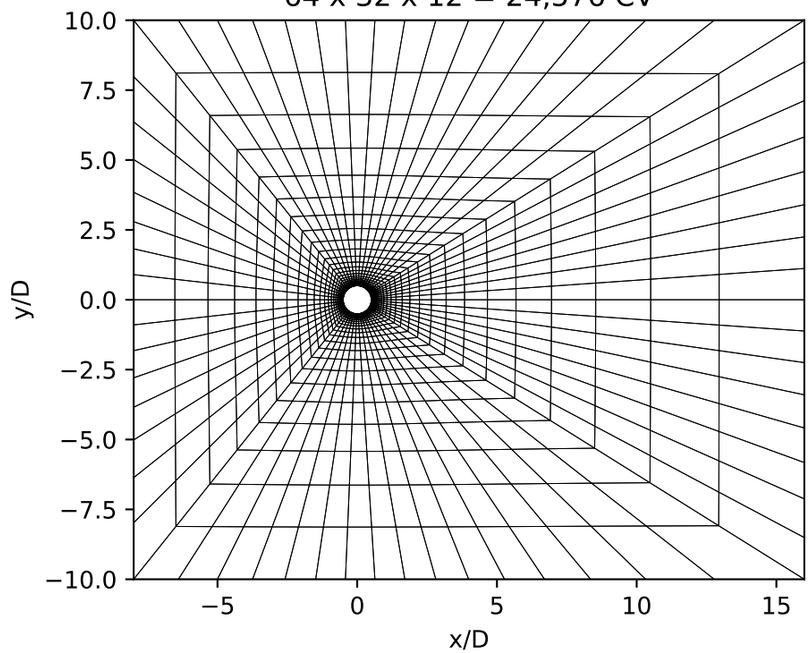
medium: actual periodic-span wall mesh



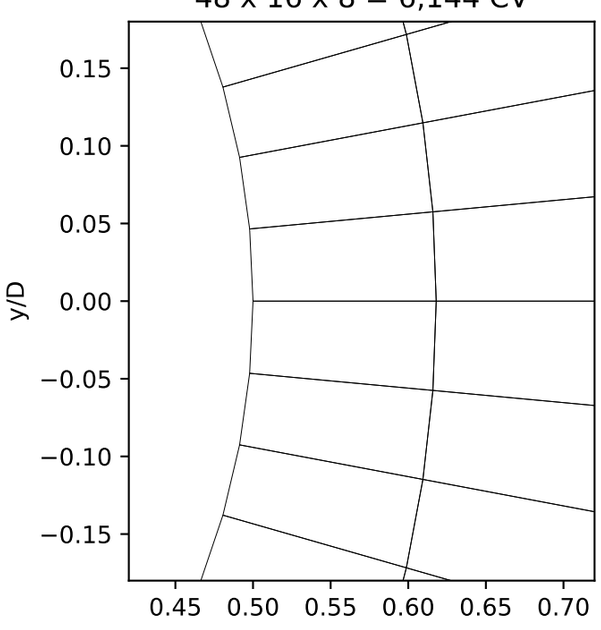
Whole midspan computational mesh
48 x 16 x 8 = 6,144 CV



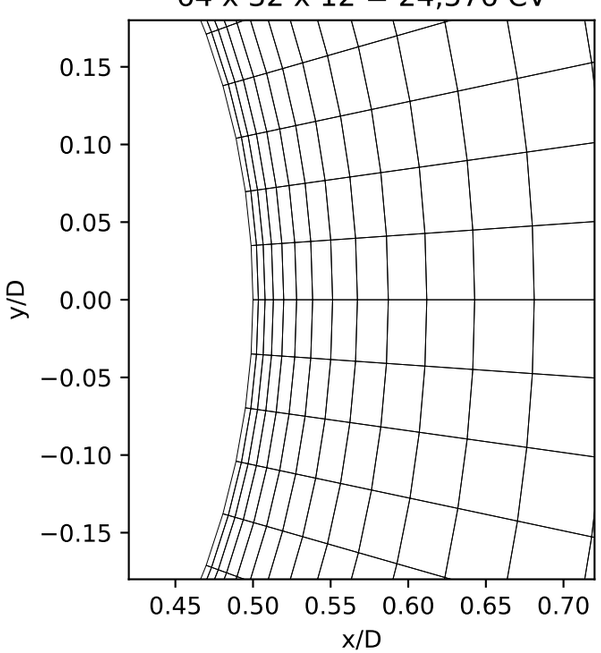
Whole midspan computational mesh
64 x 32 x 12 = 24,576 CV

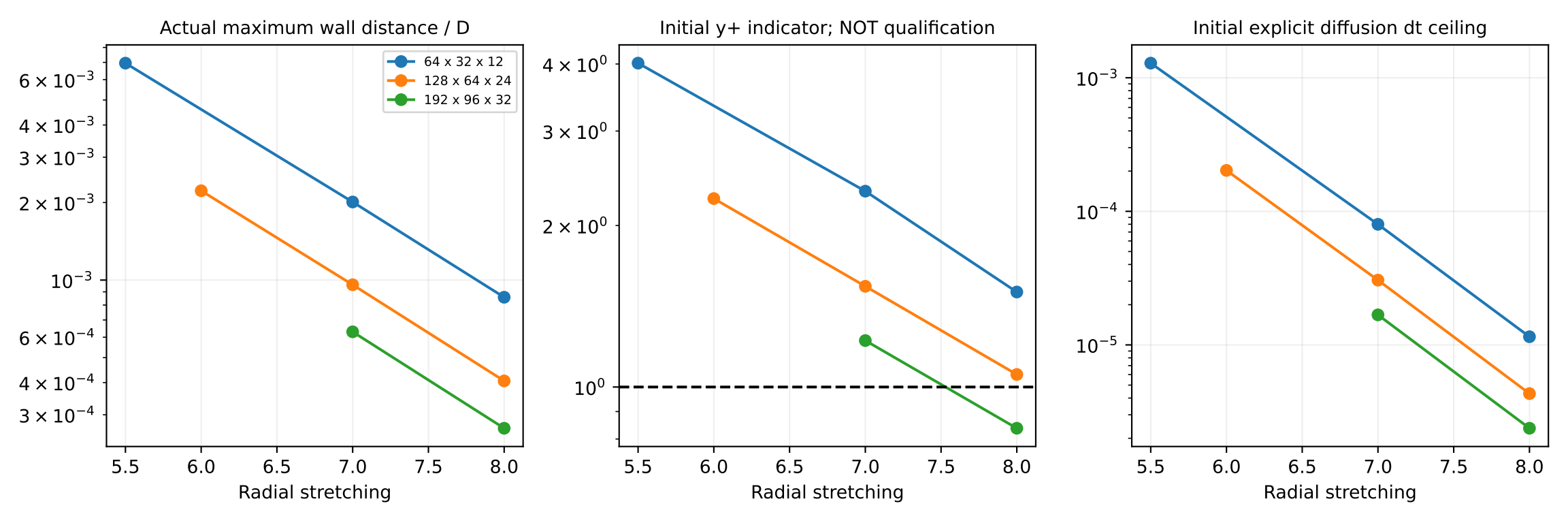


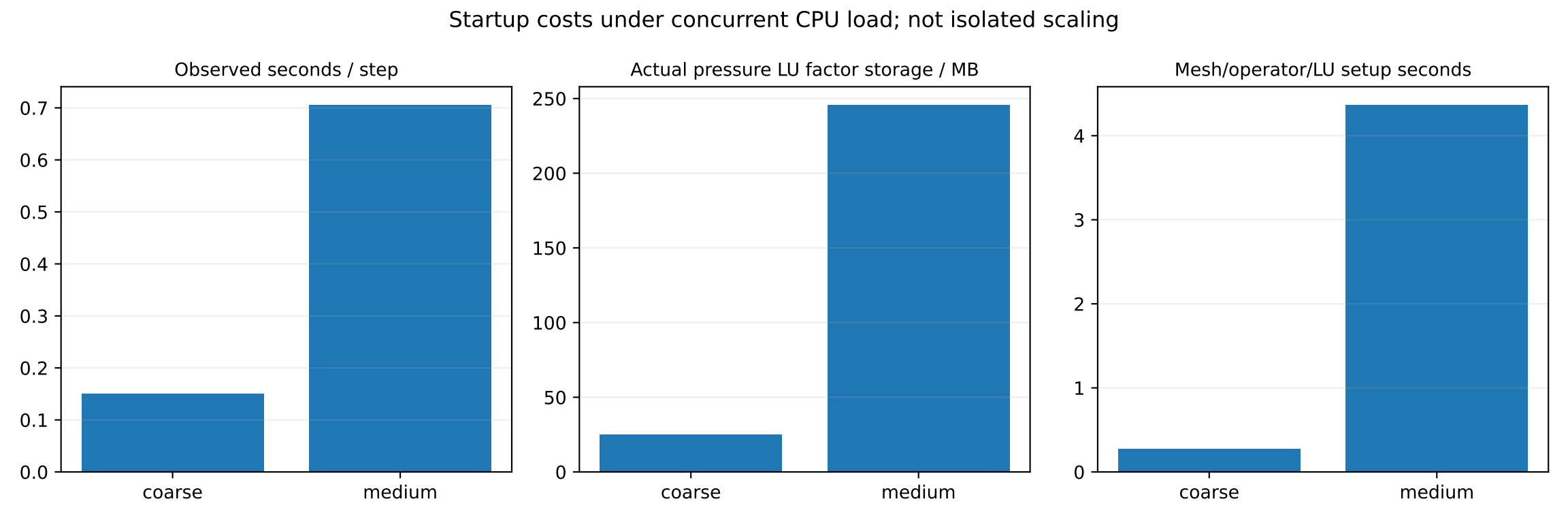
Actual near-wall cells, downstream side
48 x 16 x 8 = 6,144 CV

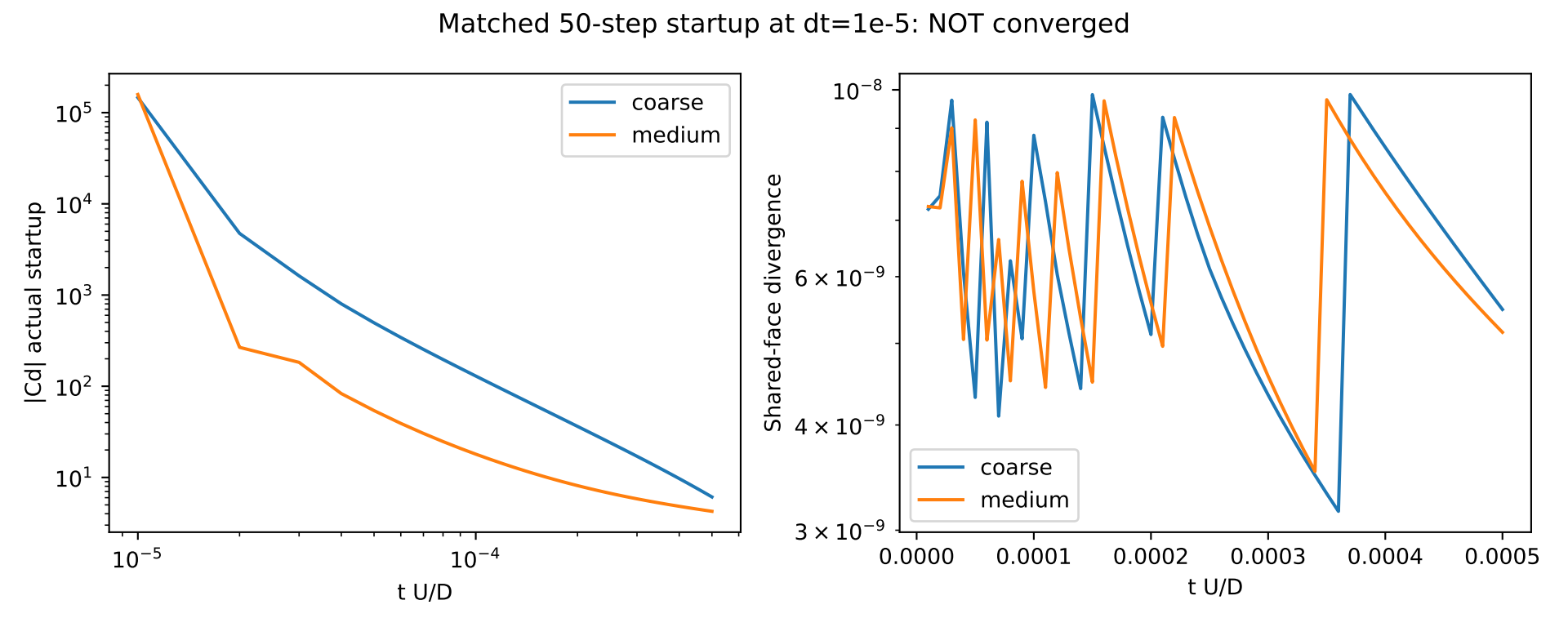


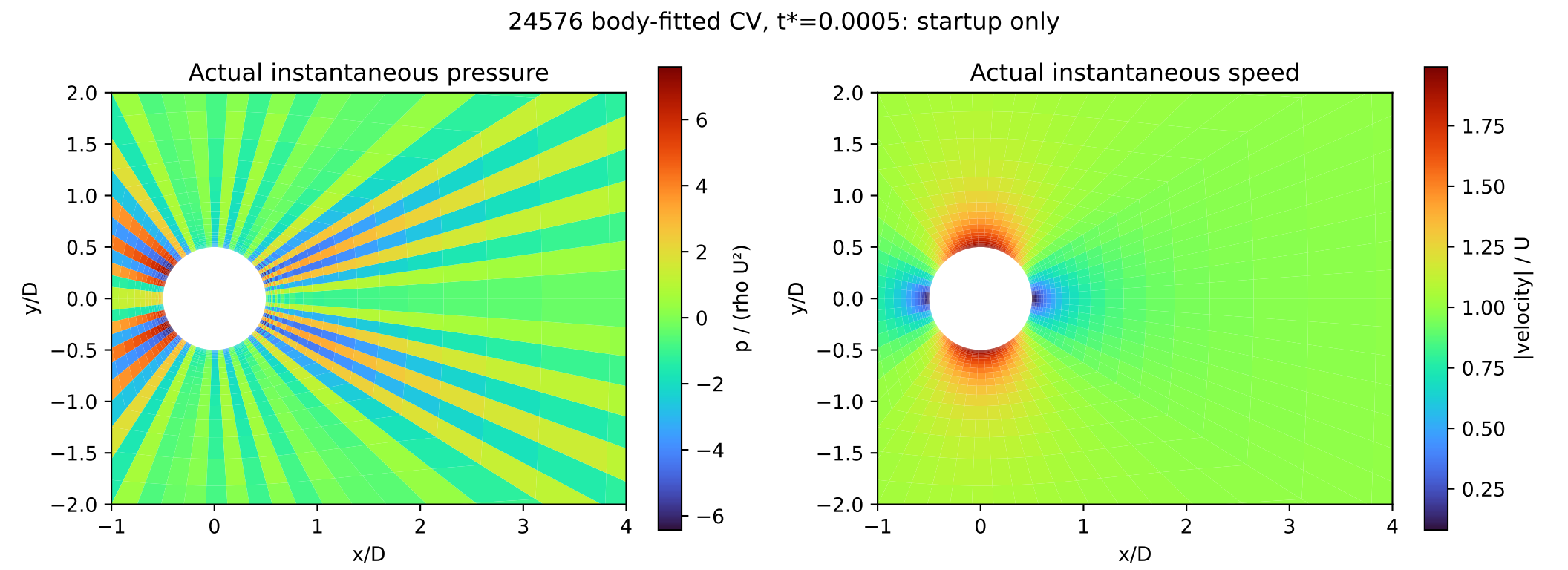
Actual near-wall cells, downstream side
64 x 32 x 12 = 24,576 CV

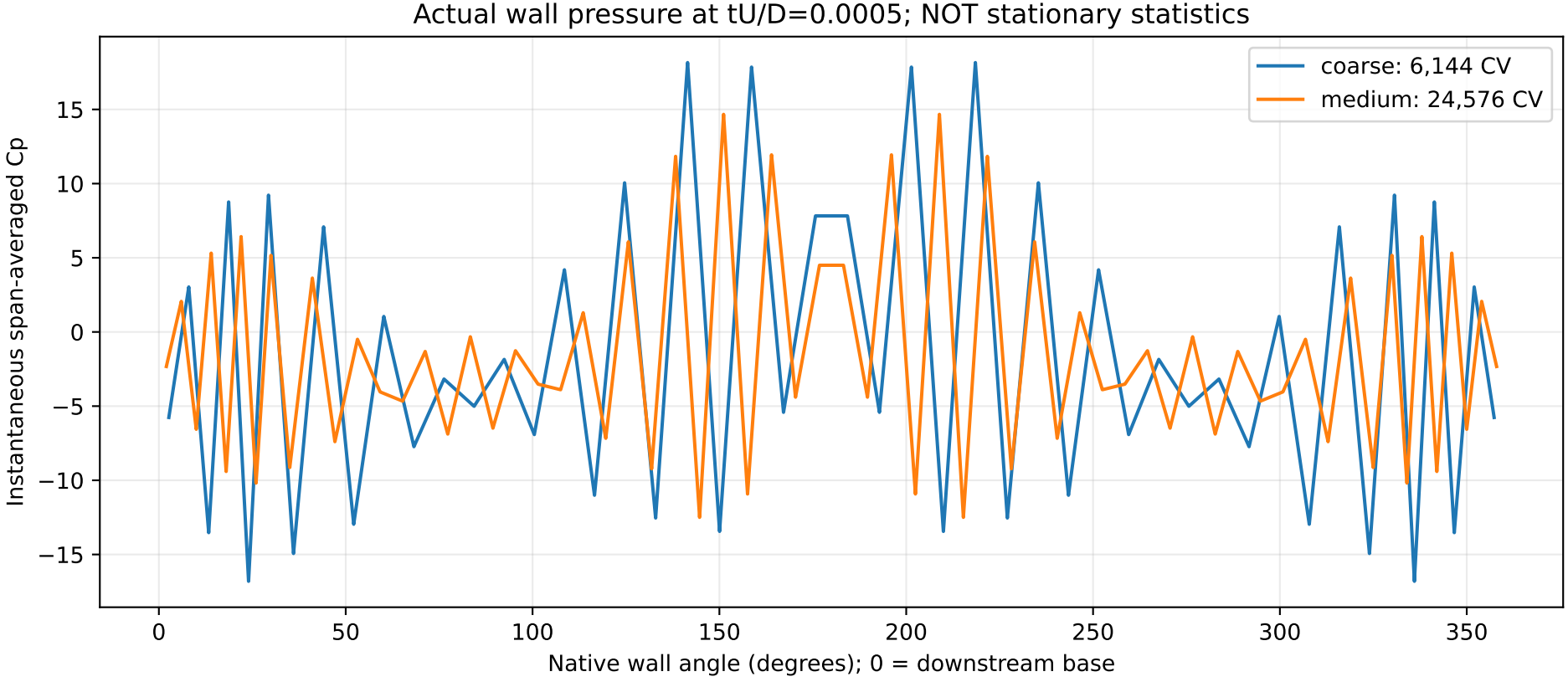



Actual maximum wall distance / D
64 x 32 x 12
128 x 64 x 24
192 x 96 x 32
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
10^{-3}
6×10^{-4}
4×10^{-4}
3×10^{-4}
5.5
6.0
6.5
7.0
7.5
8.0
Radial stretching
Initial y+ indicator; NOT qualification
4×10^{0}
3×10^{0}
2×10^{0}
10^{0}
5.5
6.0
6.5
7.0
7.5
8.0
Radial stretching
Initial explicit diffusion dt ceiling
10^{-3}
10^{-4}
10^{-5}
5.5
6.0
6.5
7.0
7.5
8.0
Radial stretching


Startup costs under concurrent CPU load; not isolated scaling
Observed seconds / step
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
coarse
medium
Actual pressure LU factor storage / MB
0
50
100
150
200
250
coarse
medium
Mesh/operator/LU setup seconds
0
1
2
3
4
coarse
medium


Matched 50-step startup at dt=1e-5: NOT converged
|Cd| actual startup
t U/D
coarse
medium
Shared-face divergence
t U/D
coarse
medium


24576 body-fitted CV, t*=0.0005: startup only
Actual instantaneous pressure
Actual instantaneous speed
y/D
x/D
p / (rho U²)
|velocity| / U


Actual wall pressure at tU/D=0.0005; NOT stationary statistics
coarse: 6,144 CV
medium: 24,576 CV
Instantaneous span-averaged Cp
15
10
5
0
−5
−10
−15
0
50
100
150
200
250
300
350
Native wall angle (degrees); 0 = downstream base